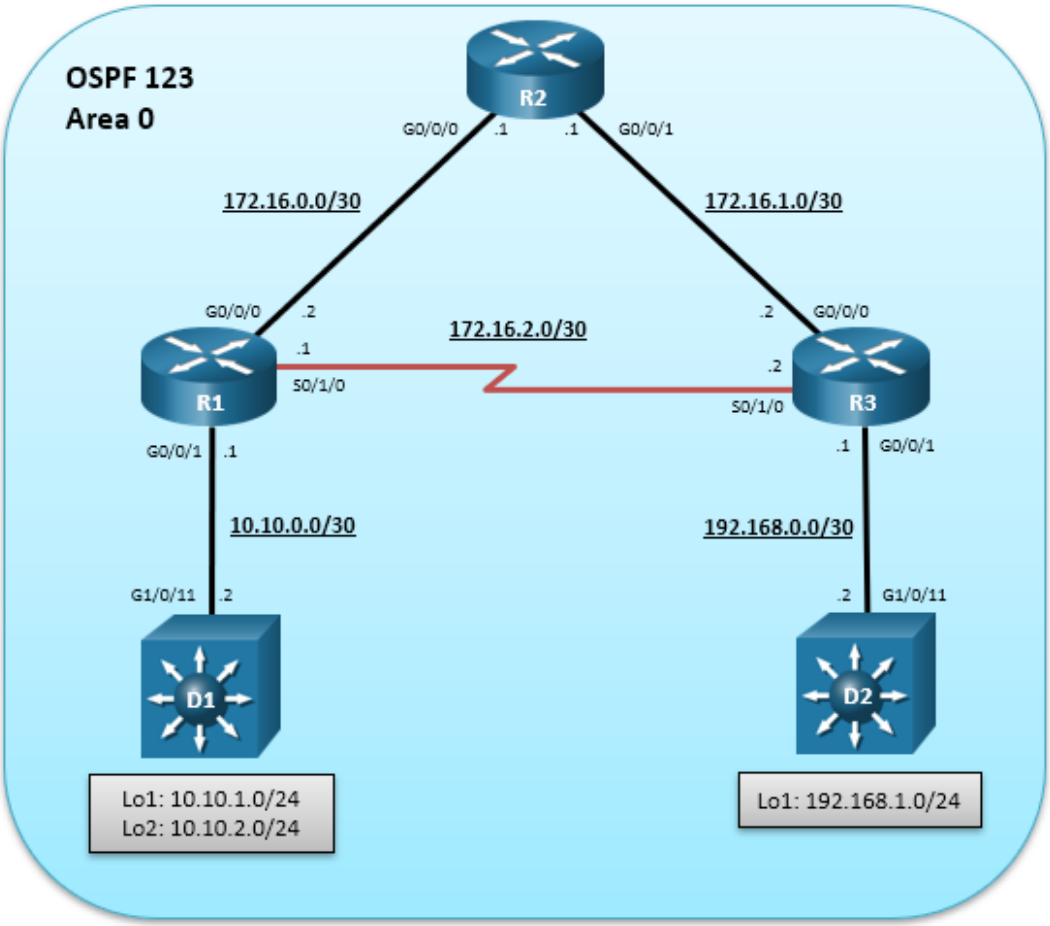
Lab - Path Control Using PBR

Topology



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask
D1	G0/0/11	10.10.0.2	255.255.255.252
	Loopback 1	10.10.1.1	255.255.255.0
	Loopback 2	10.10.2.1	255.255.255.0
R1	G0/0/0	172.16.0.2	255.255.255.252
	G0/0/1	10.10.0.1	255.255.255.252
	S0/1/0	172.16.2.1	255.255.255.252
R2	G0/0/0	172.16.0.1	255.255.255.252

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Device	Interface	IPv4 Address	Subnet Mask
	G0/0/1	172.16.1.1	255.255.255.252
R3	G0/0/0	172.16.1.2	255.255.255.252
	G0/0/1	192.168.0.1	255.255.255.252
	S0/1/0	172.16.2.2	255.255.255.252
D2	G0/0/11	192.168.0.2	255.255.255.252
	Loopback 1	192.168.1.1	255.255.255.0

Objectives

In this lab, you will learn how to use policy-based routing to influence path section.

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Configure and Verify Routing

Part 3: Configure PBR to Provide Path Control

Part 4: Configure Local PBR to Provide Path Control

Background / Scenario

In this scenario, you want to experiment with policy-based routing (PBR) to see how it is implemented and study how it could be used to influence path selection.

Your task is to connect and configure a 3 router and 2 Layer 3 switch OSPF routing domain and verify normal path selection. You will alter the traffic flow for D1 going to D2 using PBR. You will also use Cisco IOS IP SLA with PBR to achieve dynamic path control. Finally, you will configure PBR for traffic originating locally on router R1.

Note: This lab is an exercise in configuring distribute lists and redistribution using ALCs, prefix lists, and route maps.

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.5 (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.5 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the devices have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.5c universal image or comparable)
- 2 Switches (Catalyst 9300L with Cisco IOS XE Release 17.12.5 universalk9 image).
- 1 PC (Choice of operating system with terminal emulation program installed)

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Instructions

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for each device.

a. Console into each device, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

Router R1

```
hostname R1
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
banner motd # This is R1, Path Control Using PBR #
interface G0/0/0
 description Connection to R2
 ip add 172.16.0.2 255.255.255.252
 no shut
 exit
interface S0/1/0
 description Serial Connection to R3
 ip add 172.16.2.1 255.255.255.252
 no shut
 exit
interface G0/0/1
 description Connection to D1
 ip add 10.10.0.1 255.255.255.252
 no shut
 exit
```

Router R2

```
hostname R2
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
exit
banner motd # This is R2, Path Control Using PBR #
interface G0/0/0
```

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```
 description Connection to R1
 ip add 172.16.0.1 255.255.255.252
 no shut
exit
interface GigabitEthernet0/0/1
 description Connection to R3
 ip address 172.16.1.1 255.255.255.252
 no shut
exit
```

Router R3

```
hostname R3
no ip domain lookup
line con 0
 logging sync
 exec-time 0 0
 exit
banner motd # This is R3, Path Control Using PBR #
interface G0/0/0
 description Connection to R2
 ip add 172.16.1.2 255.255.255.252
 no shut
 exit
interface S0/1/0
 description Serial Connection to R1
 ip add 172.16.2.2 255.255.255.252
 no shut
 exit
interface G0/0/1
 description Connection to D2
 ip add 192.168.0.1 255.255.255.252
 no shut
 exit
```

Switch D1

```
hostname D1
no ip domain lookup
line con 0
exec-timeout 0 0
logging synchronous
exit
banner motd # This is D1, Path Control Using PBR #
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
interface G1/0/11
 no switchport
```

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```
 description Connects to R1
 ip address 10.10.0.2 255.255.255.252
 no shut
 exit
interface Loopback 1
 description Interface simulates network
 ip ospf network point-to-point
 ip address 10.10.1.1 255.255.255.0
 exit
interface Loopback 2
 description Interface simulates network
 ip ospf network point-to-point
 ip address 10.10.2.1 255.255.255.0
 exit
```

Switch D2

```
hostname D2
no ip domain lookup
line con 0
logging sync
exec-time 0 0
exit
banner motd # This is D2, Path Control Using PBR #
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
interface G1/0/11
 no switchport
 description Connects to R3
 ip address 192.168.0.2 255.255.255.252
 no shut
 exit
interface Loopback 1
 description Interface simulates network
 ip ospf network point-to-point
 ip address 192.168.1.1 255.255.255.0
 exit
```

b. Save the running configuration to startup-config.

Part 2: Configure and Verify Routing

In Part 2, you will implement OSPF routing for the routing domain and verify end to end routing.

Step 1: Configure Routing.

In this step, you will configure OSPF.

a. On D1, advertise the connected networks using OSPF process ID 123. Also assign D1 the router ID of 1.1.1.2 and set the reference bandwidth to recognize Gigabit Ethernet interfaces.

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```
D1(config)# ip routing
D1(config)# router ospf 123
D1(config-router)# router-id 1.1.1.2
D1(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
D1(config-router)# network 10.10.0.0 0.0.0.3 area 0
D1(config-router)# network 10.10.1.0 0.0.0.255 area 0
D1(config-router)# network 10.10.2.0 0.0.0.255 area 0
D1(config-router)# end
```

b. On R1, advertise the connected networks using OSPF process ID 123. Also assign R1 the router ID of 1.1.1.1 and set the reference bandwidth to recognize Gigabit Ethernet interfaces.

```
R1(config)# router ospf 123
R1(config-router)# router-id 1.1.1.1
R1(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R1(config-router)# network 10.10.0.0 0.0.0.3 area 0
R1(config-router)# network 172.16.0.0 0.0.0.3 area 0
R1(config-router)# network 172.16.2.0 0.0.0.3 area 0
R1(config-router)# end
R1#
*Feb 19 17:00:40.661: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.2 on GigabitEthernet0/0/1
from LOADING to FULL, Loading Done
```

c. On R2, advertise the connected networks using OSPF process ID 123. Also assign R2 the router ID of 2.2.2.1 and set the reference bandwidth to recognize Gigabit Ethernet interfaces.

```
R2(config)# router ospf 123
R2(config-router)# router-id 2.2.2.1
R2(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
*Feb 19 17:02:34.016: %OSPF-6-DFT_OPT: Protocol timers for fast convergence
areEnabled.172.
R2(config-router)# network 172.16.0.0 0.0.0.3 area 0
R2(config-router)# network 172.16.1.0 0.0.0.3 area 0
R2(config-router)# end
R2#
*Feb 19 17:02:42.460: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on GigabitEthernet0/0/0
from LOADING to FULL, Loading Done
```

d. On R3, advertise the connected networks using OSPF process ID 123. Also assign R3 the router ID of 3.3.3.1 and set the reference bandwidth to recognize Gigabit Ethernet interfaces.

```
R3(config)# router ospf 123
R3(config-router)# router-id 3.3.3.1
R3(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

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```
R3(config-router)# network 192.168.0.0 0.0.0.3 area 0
R3(config-router)# network 172.16.1.0 0.0.0.3 area 0
R3(config-router)# network 172.16.2.0 0.0.0.3 area 0
R3(config-router)# end
R3#
*Feb 19 17:03:56.362: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on Serial0/1/0 from
LOADING to FULL, Loading Done
R3#
*Feb 19 17:09:38.978: %OSPF-5-ADJCHG: Process 123, Nbr 2.2.2.1 on GigabitEthernet0/0/0
from LOADING to FULL, Loading Done
```

e. On D2, advertise the connected networks using OSPF process ID 123. Also assign D2 the router ID of 3.3.3.2 and set the reference bandwidth to recognize Gigabit Ethernet interfaces.

```
D2(config)# ip routing
D2(config)# router ospf 123
D2(config-router)# router-id 3.3.3.2
D2(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
D2(config-router)# network 192.168.0.0 0.0.0.3 area 0
D2(config-router)# network 192.168.1.0 0.0.0.255 area 0
D2(config-router)# end
D2#
*Feb 19 17:29:46.627: %OSPF-5-ADJCHG: Process 123, Nbr 3.3.3.1 on
GigabitEthernet1/0/11 from LOADING to FULL, Loading Done
```

Step 2: Verify OSPF routing

a. Before configuring PBR, verify the current routing table on all devices. All routing tables look accurate.

```
D1# show ip route ospf | begin Gateway
Gateway of last resort is not set

      172.16.0.0/30 is subnetted, 3 subnets
O        172.16.0.0 [110/2] via 10.10.0.1, 00:11:36, GigabitEthernet1/0/11
O        172.16.1.0 [110/3] via 10.10.0.1, 00:10:10, GigabitEthernet1/0/11
O        172.16.2.0 [110/497] via 10.10.0.1, 00:11:36, GigabitEthernet1/0/11
      192.168.0.0/30 is subnetted, 1 subnets
O        192.168.0.0 [110/4] via 10.10.0.1, 00:09:57, GigabitEthernet1/0/11
O     192.168.1.0/24 [110/14] via 10.10.0.1, 00:09:57, GigabitEthernet1/0/11

D2# show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.10.0.0/30 [110/4] via 192.168.0.1, 00:03:55, GigabitEthernet1/0/11
O        10.10.1.0/24 [110/5] via 192.168.0.1, 00:03:55, GigabitEthernet1/0/11
O        10.10.2.0/24 [110/5] via 192.168.0.1, 00:03:55, GigabitEthernet1/0/11
      172.16.0.0/30 is subnetted, 3 subnets
O        172.16.0.0 [110/3] via 192.168.0.1, 00:03:55, GigabitEthernet1/0/11
```

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```
O        172.16.1.0 [110/2] via 192.168.0.1, 00:03:55, GigabitEthernet1/0/11
O        172.16.2.0 [110/499] via 192.168.0.1, 00:03:55, GigabitEthernet1/0/1

R1# show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
O        10.10.1.0/24 [110/11] via 10.10.0.2, 00:12:21, GigabitEthernet0/0/1
O        10.10.2.0/24 [110/11] via 10.10.0.2, 00:12:21, GigabitEthernet0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O        172.16.1.0/30 [110/2] via 172.16.0.1, 00:10:55, GigabitEthernet0/0/0
      192.168.0.0/30 is subnetted, 1 subnets
O        192.168.0.0 [110/3] via 172.16.0.1, 00:10:42, GigabitEthernet0/0/0
O     192.168.1.0/24 [110/13] via 172.16.0.1, 00:10:42, GigabitEthernet0/0/0

R2# show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
O        10.10.0.0/30 [110/2] via 172.16.0.2, 00:12:26, GigabitEthernet0/0/0
O        10.10.1.0/24 [110/12] via 172.16.0.2, 00:12:26, GigabitEthernet0/0/0
O        10.10.2.0/24 [110/12] via 172.16.0.2, 00:12:26, GigabitEthernet0/0/0
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O        172.16.2.0/30
           [110/497] via 172.16.0.2, 00:12:26, GigabitEthernet0/0/0
      192.168.0.0/30 is subnetted, 1 subnets
O        192.168.0.0 [110/2] via 172.16.1.2, 00:11:22, GigabitEthernet0/0/1
O     192.168.1.0/24 [110/12] via 172.16.1.2, 00:11:22, GigabitEthernet0/0/1

R3# show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.10.0.0/30 [110/3] via 172.16.1.1, 00:12:56, GigabitEthernet0/0/0
O        10.10.1.0/24 [110/13] via 172.16.1.1, 00:12:56, GigabitEthernet0/0/0
O        10.10.2.0/24 [110/13] via 172.16.1.1, 00:12:56, GigabitEthernet0/0/0
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O        172.16.0.0/30 [110/2] via 172.16.1.1, 00:12:56, GigabitEthernet0/0/0
O     192.168.1.0/24 [110/11] via 192.168.0.2, 00:12:44, GigabitEthernet0/0/1
```

Step 3: Verify end-to-end connectivity and path taken

a. From any device, verify connectivity to all configured destinations using the following TCL script. All pings should be successful. Troubleshoot if necessary.

```
tclsh

foreach address {
10.10.0.1
10.10.0.2
```

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```
10.10.1.1
10.10.2.1
172.16.0.1
172.16.0.2
172.16.1.1
172.16.1.2
172.16.2.1
172.16.2.2
192.168.0.1
192.168.0.2
192.168.1.1
} { ping $address }
```

b. Enter **tclquit** to exit tcl mode.

c. On D1, ping the D2 Loopback interface 192.168.1.1 address from the Lo1 interface as shown. The pings should be successful.

```
D1# ping 192.168.1.1 source 10.10.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
```

d. Next, identify the path taken to D2 Lo1 interface using the **traceroute** command as shown. Notice that the path taken for the packets sourced from the D1 Lo1 LAN is going through R1 --> R2 --> R3 --> D2.

```
D1# traceroute 192.168.1.1 source 10.10.1.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.0.1 2 msec 2 msec 2 msec
  2 172.16.0.1 2 msec 2 msec 2 msec
  3 172.16.1.2 2 msec 2 msec 2 msec
  4 192.168.0.2 3 msec *  3 msec
```

Why is the path not taking the shorter R1 --> R3 --> D2 path?

e. Now **ping** and **traceroute** the D2 Lo1 interface from the D1 Loopback 2 interface as shown. It is also taking the same path.

```
D1# ping 192.168.1.1 source lo 2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms

D1# traceroute 192.168.1.1 source lo 2
Type escape sequence to abort.
Tracing the route to 192.168.1.1
```

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```
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.0.1 2 msec 2 msec 1 msec
  2 172.16.0.1 1 msec 2 msec 1 msec
  3 172.16.1.2 2 msec 2 msec 2 msec
  4 192.168.0.2 3 msec *  3 msec
```

f. Display the OSPF routes in the routing table of R1. R1 forwards all packets destined to the 192.168.1.0/24 network out of its G0/0/0 interface to R2.

```
R1# show ip route ospf | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
O        10.10.1.0/24 [110/2] via 10.10.0.2, 00:19:56, GigabitEthernet0/0/1
O        10.10.2.0/24 [110/2] via 10.10.0.2, 00:19:56, GigabitEthernet0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O        172.16.1.0/30 [110/2] via 172.16.0.1, 00:18:50, GigabitEthernet0/0/0
      192.168.0.0/30 is subnetted, 1 subnets
O        192.168.0.0 [110/3] via 172.16.0.1, 00:18:37, GigabitEthernet0/0/0
O     192.168.1.0/24 [110/4] via 172.16.0.1, 00:18:33, GigabitEthernet0/0/0
```

g. Display how R1 learned about the 192.168.1.0 network. R1 learned of the network from R2 (i.e., 172.16.0.1) who originally learned it from D2 (i.e., 3.3.3.2).

```
R1# show ip route 192.168.1.0
Routing entry for 192.168.1.0/24
  Known via "ospf 123", distance 110, metric 4, type intra area
  Last update from 172.16.0.1 on GigabitEthernet0/0/0, 00:20:27 ago
  Routing Descriptor Blocks:
  * 172.16.0.1, from 3.3.3.2, 00:20:27 ago, via GigabitEthernet0/0/0
      Route metric is 4, traffic share count is 1
```

Part 3: Configure PBR to Provide Path Control

Recall that route maps can be used for:

- **Redistribution** - Route maps provide more options and flexibility to the **redistribute** command.
- **Policy-based routing (PBR)** - PBR allows an administrator to define routing policy other than basic destination-based routing using the routing table. The route map is applied to an interface using the **ip policy route-map** interface configuration command.
- **BGP** - Route maps are the primary tools for implementing BGP policy and allow an administrator to do path control and provide sophisticated manipulation of BGP path attributes. The route map is applied using the BGP **neighbor** router configuration command.

In this part, you will use PBR to configure source-based IP routing. Specifically, you will override the default IP routing decision based on the OSPF-acquired routing information for selected IP source-to-destination flows and apply a different next-hop router.

Recall that routers normally forward packets to destination addresses based on information in their routing table. By using PBR, you can implement policies that selectively cause packets to take different paths based on source address, protocol type, or application type. Therefore, PBR overrides the router's normal routing behavior.

Configuring PBR involves configuring a route map with **match** and **set** commands and then applying the route map to the interface.

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The steps required to implement path control include the following:

- Choose the path control tool to use. Path control tools manipulate or bypass the IP routing table. For PBR, **route-map** commands are used.
- Implement the traffic-matching configuration, specifying which traffic will be manipulated. The **match** commands are used within route maps.
- Define the action for the matched traffic using **set** commands within route maps.
- Apply the route map to incoming traffic.

Step 1: Configure PBR on R1.

As a test, you will configure the following policy on router R1:

- All traffic sourced from D1 Lo1 LAN must take the R1 --> R2 --> R3 --> D2 path.
- All traffic sourced from D1 Lo2 LAN must take the R1 --> R3 --> D2 path.

a. On R1, create a standard named ACL called **Lo2-ACL** to identify the D1 Loopback 2 (i.e., 10.10.2.0/24) LAN.

```
R1(config)# ip access-list standard Lo2-ACL
R1(config-std-nacl)# remark ACL matches D1 Lo2 traffic
R1(config-std-nacl)# permit 10.10.2.0 0.0.0.255
R1(config-std-nacl)# exit
```

b. Create a route map called **R1-to-R3** that matches Lo2-ACL and sets the next-hop interface to the R3 serial 0/1/0 interface.

```
R1(config)# route-map R1-to-R3 permit
R1(config-route-map)# description RM to forward Lo2 traffic to R3
R1(config-route-map)# match ip address Lo2-ACL
R1(config-route-map)# set ip next-hop 172.16.2.2
R1(config-route-map)# exit
```

c. Apply the R1-to-R3 route map to the G0/0/1 interface using the **ip policy route-map** command.

```
R1(config)# interface g0/0/1
R1(config-if)# ip policy route-map R1-to-R3
R1(config-if)# end
```

d. On R1, display the policy and matches using the **show route-map** command.

```
R1# show route-map
route-map R1-to-R3, permit, sequence 10
  Match clauses:
    ip address (access-lists): Lo2-ACL
  Set clauses:
    ip next-hop 172.16.2.2
  Policy routing matches: 0 packets, 0 bytes
```

Note: There are currently no matches because no packets matching the ACL have passed through R1 G0/0/1.

e. On R1, verify that the R1-to-R3 route map has been applied to the G0/0/1 interface.

```
R1# show ip policy
Interface      Route map
Gi0/0/1        R1-to-R3
```

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Step 2: Test the policy.

Now you are ready to test the policy configured on R1.

a. From D1, test the policy with the **traceroute** command, using D1 Lo1 interface as the source network.

```
D1# traceroute 192.168.1.1 source lo 1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.0.1 2 msec 1 msec 1 msec
  2 172.16.0.1 2 msec 2 msec 2 msec
  3 172.16.1.2 2 msec 4 msec 2 msec
  4 192.168.0.2 3 msec *  3 msec
```

Notice the path taken for the packet sourced from D1 Lo 1 LAN A is still going through R1 --> R2 --> R3 --> D2.

Why is the traceroute traffic not using the R3 --> R1 path as specified in the R1-to-R3 policy?

b. Now test the policy with the **traceroute** command, using D1 Lo2 interface as the source network. Now the path taken for the packet sourced from D1 Lo 2 LAN is R1 --> R3 --> D2, as expected.

```
D1# traceroute 192.168.1.1 source lo 2
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.0.1 2 msec 2 msec 1 msec
  2 172.16.2.2 3 msec 2 msec 3 msec
  3 192.168.0.2 3 msec *  4 msec
```

c. On R1, display the policy and matches using the **show route-map** command.

Note: There are now matches to the policy because packets matching the ACL have passed through R1 G0/0/1 interface. The number of packet and bytes may differ in your implementation.

```
R1# show route-map
route-map R1-to-R3, permit, sequence 10
  Match clauses:
    ip address (access-lists): Lo2-ACL
  Set clauses:
    ip next-hop 172.16.2.2
  Policy routing matches: 5 packets, 210 bytes
```

Part 4: Configure Local PBR to Provide Path Control

How would you policy route packets generated by a router? The answer is to configure Local PBR.

Local PBR is a feature to policy route locally generated traffic. Local PBR policies are applied to the router with the **ip local policy route-map** global config command.

In this part, you will configure R1 to policy route all router generated traffic over the R1 to R3 link.

Step 1: Configure Local PBR on R1.

a. Verify the path that R1 currently takes without local PBR configured. R1 sends traffic to R2 then R3 and finally D2 as expected.

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```
R1# traceroute 192.168.1.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.0.1 1 msec 1 msec 1 msec
  2 172.16.1.2 1 msec 2 msec 1 msec
  3 192.168.0.2 2 msec *  2 msec
```

b. On R1, create a named extended ACL called **R1-TRAFFIC** which matches all IP generated packets from R1 and destined to the D2 192.162.1.0/24 network.

```
R1(config)# ip access-list extended R1-TRAFFIC
R1(config-ext-nacl)# permit ip any 192.168.1.0 0.0.0.255
R1(config-ext-nacl)# exit
```

c. On R1, create a route map called **LOCAL-PBR** that permits traffic matching the R1-TRAFFIC ACL and redirects it to the R3 172.16.2.2 interface.

```
R1(config)# route-map LOCAL-PBR permit
R1(config-route-map)# match ip address R1-TRAFFIC
R1(config-route-map)# set ip next-hop 172.16.2.2
R1(config-route-map)# exit
```

d. Create a local PBR policy that matches the LOCAL-PBR route map.

```
R1(config)# ip local policy route-map LOCAL-PBR
R1(config)# exit
```

Step 2: Test Local PBR on R1.

a. Verify the path taken by R1 to reach the 192.168.1.0/24 LAN. The traffic generated by R1 and going to 192.168.1.0/24 is now policy routed directly to R3 (i.e., 172.16.2.2).

```
R1# traceroute 192.168.1.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.2.2 2 msec 1 msec 2 msec
  2 192.168.0.2 2 msec *  2 msec
```

b. Verify the path taken by R1 to reach other networks. The traffic takes the normal OSPF generated path and is not policy routed.

```
R1# traceroute 192.168.0.2
Type escape sequence to abort.
Tracing the route to 192.168.0.2
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.0.1 1 msec 1 msec 1 msec
  2 172.16.1.2 1 msec 2 msec 1 msec
  3 192.168.0.2 4 msec *  2 msec
```

c. Verify the route-map counters. The local PBR policy has matched packets.

Note: The number of packets and bytes may differ in your implementation.

```
R1# show route-map
route-map R1-to-R3, permit, sequence 10
```

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```
    Match clauses:
      ip address (access-lists): Lo2-ACL
    Set clauses:
      ip next-hop 172.16.2.2
    Policy routing matches: 6 packets, 252 bytes
route-map LOCAL-PBR, permit, sequence 10
    Match clauses:
      ip address (access-lists): ICMP-TRAFFIC
    Set clauses:
      ip next-hop 172.16.2.2
    Policy routing matches: 32 packets, 2384 bytes
```

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